Infant–Caregiver Co-Regulation Is Bidirectional

Co-regulation is a bidirectional, context-sensitive process in which infant and caregiver organize arousal together; synchrony is not uniformly beneficial, and nurturing care is an ecology, not a maternal purity test.

ODERATE-EXISTENCE; LOW-TO-MODERATE-MECHANISM | STATUS: DRAFT | DOMAIN: REGULATION, ATTACHMENT | REPLICATION:

WHAT WE KNOW (Narrowest Supported Conclusion)

Human infants have limited capacity to stabilize arousal, temperature, feeding, sleep, and distress alone. Caregivers help organize those states through contingent, repeated interaction. The process is bidirectional: infant signals shape the adult; adult response shapes the infant. WHO/UNICEF/World Bank Nurturing Care treats responsive caregiving as one component of an ecology that also includes health, nutrition, safety, and early learning.

EVIDENCE AND CAUSAL ASSESSMENT

Feldman (2012) models affiliative bonds as time-locked coordination of physiology and interactive behavior; it is a theoretical synthesis of observational and psychophysiological work, not an RCT of "synchrony training." The Nurturing Care framework is a policy synthesis, not a causal estimate of ordinary parenting differences. Existing cards already cover sensitivity-training RCTs (responsive-caregiving-attachment-interventions.md) and sleep-training RCTs (infant-sleep-training.md); those claims are not restated here.

EFFECT SIZE AND PERSISTENCE

There is no single effect size for "co-regulation." Immediate changes in arousal during an interaction are more defensible than permanent biological programming. Long-run executive-function or personality claims require separate identification.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (bidirectional dynamic interaction framework, not an observational dosage law).

Infant distress can move caregiver physiology (child-to-parent effects). Matching can be engagement, recovery, or stress contagion depending on timing and system. Material conditions (leave, housing, health services) enable or constrain responsiveness; treating the parent as a sole exposure is a specification error.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Responding to crying is not automatically "spoiling," and waiting is not automatically "teaching independence." Synchrony is not a purity test. Nurturing care is not a demand that one adult regulate every state perfectly.

WHAT ACTION IS REASONABLY JUSTIFIED

Support contingent, developmentally appropriate interaction and the conditions that make it possible. Promises of biological optimization, or using this card to rewrite the sleep and attachment cards, are not justified.

VERIFIED SOURCES & DOI IDENTIFIERS

- WHO, UNICEF & World Bank Group (2018), Nurturing care for early childhood development
- Feldman (2012), Parenting: Science and Practice